

UNDECIDABLE DIOPHANTINE EQUATIONS

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Corrected transcription, continuously revised: every discrepancy between the printed announcement and this edition is explained in `Papers/1980/jones1980_editorial_notes.md`, and a dated log of revisions and checks is `Papers/EDITORIAL_NOTES.md`. Mathematical and bibliographic emendations are documented in the accompanying editorial notes. The results and bounds are those announced in the 1980 article, not a survey of later work. The margin numbers [859]–[862] mark the lines on which the pages of the original printing begin.

In 1900 Hilbert asked for an algorithm to decide the solvability of all diophantine equations, [859] $P(x_1, \dots, x_\nu) = 0$, where P is a polynomial with integer coefficients. In special cases of Hilbert's tenth problem, such algorithms are known. Siegel [7] gives an algorithm for all polynomials $P(x_1, \dots, x_\nu)$ of degree ≤ 2 . From the work of A. Baker [1] we know that there is also a decision procedure for the case of homogeneous polynomials in two variables, $P(x, y) = c$.

An essential step toward the eventual negative solution of the entire (unrestricted) form of Hilbert's tenth problem was taken in 1961 by Julia Robinson, Martin Davis and Hilary Putnam [2]. They proved that every recursively enumerable set W can be represented in exponential diophantine form

$$x \in W \iff \exists x_1, \dots, x_\mu P(x, x_1, \dots, x_\mu, 2^{x_1}, \dots, 2^{x_\mu}) = 0,$$

where P is a polynomial with integer coefficients and $x_1, \dots, x_\mu$ range over positive integers.

In 1970 Ju. V. Matijasevič [4] proved that the exponential relation $y = 2^x$ is diophantine and hence that every r.e. set W can be represented in polynomial (diophantine) form

$$x \in W \iff \exists x_1, \dots, x_\nu P(x, x_1, \dots, x_\nu) = 0,$$

where the unknowns $x_1, \dots, x_\nu$ range over positive integers.

Since there exist r.e. nonrecursive sets, Matijasevič's theorem implies the undecidability of Hilbert's tenth problem. There is no algorithm to decide whether an arbitrary diophantine equation has a solution.

Matijasevič's theorem implies also the existence of particular undecidable families of diophantine equations. In fact there must exist universal diophantine equations, polynomial analogues of the universal Turing machine. This follows from the well-known fact that the r.e. sets $W_1, W_2, \dots$ can be listed in such a way that the binary relation $x \in W_v$ is r.e. Hence by [2, 4] there exists a universal polynomial $U(x, v, x_1, \dots, x_\nu)$ with the property

$$x \in W_v \iff \exists x_1, \dots, x_\nu U(x, v, x_1, \dots, x_\nu) = 0.$$

Thus a single polynomial, of a fixed degree and a fixed number of unknowns, can define every r.e. set by mere change of a parameter v . The existence of such a universal polynomial follows immediately [860] from [2, 4]. However, the construction of an actual example is a different matter. The first example to be specifically written down is given in [3]. More recently the author has constructed further examples. These are given below, without proof. (The proof uses the code idea of [6] together with new methods of Matijasevič based on E. Kummer's theorem about divisibility of binomial coefficients by prime powers.)

Editorial index convention. In Theorems 1–3 the triples $\langle z, u, y \rangle$ are admissible polynomial codes: the triples constructed in (4.1) of Jones’s 1982 *Universal Diophantine Equation*, with the normalization specified in its §3. This is the class of indices for which the common equivalences are proved. The precise coefficient convention is recorded in the editorial notes (entry 1980-08); an arbitrary unrelated positive triple is not automatically such a code.

Theorem 1. *In order that $x \in W_{\langle z, u, y \rangle}$, it is necessary and sufficient that the following system of equations have a solution in positive integers.*

$$\begin{aligned} elg^2 + \alpha &= (b - xy)q^2, & q &= b^{560}, \\ \lambda + q^4 &= 1 + \lambda b^5, & \theta + 2z &= b^5, \\ \ell &= u + t\theta, & e &= y + m\theta, & b &= 2^w, \\ \binom{g + q^3 - 1 - b\ell + \ell}{g} \binom{\theta\lambda + e + \ell q^2}{\theta\lambda} \\ &\cdot \binom{b^5q - 2q + 2(e - z\lambda)(1 + xb^5 + g)^4 + b^5\lambda(1 + q^4)}{b^5q - 2q} = 2\eta + 1. \end{aligned}$$

Here we consider the r.e. sets to be indexed by three indices, z, u, y , instead of the usual one. This is an inessential restriction. To use indexing with a single parameter v , one need only add to our equations the equation

$$v = ((zuy)^2 + u)^2 + y$$

and regard z, u, y as unknowns instead of parameters.¹

As stated above, our system of equations has 12 unknowns, specifically $b, e, g, \ell, m, q, t, w, \alpha, \eta, \theta, \lambda$, and four parameters, z, u, y, x . An exponential equation, $b = 2^w$, appears, and the last line is a product of three binomial coefficients. In the next system the number of binomial coefficients is reduced to one.

Theorem 2. *In order that $x \in W_{\langle z, u, y \rangle}$, it is necessary and sufficient that the following system of equations have a solution in positive integers.*

$$\begin{aligned} elg^2 + \alpha &= (b - xy)q^2, & q &= b^{560}, \\ \lambda + q^4 &= 1 + \lambda b^5, & \theta + 2z &= b^5, \\ \ell &= u + t\theta, & e &= y + m\theta, & b &= 2^w, & n &= q^{16}, \\ r &= \left[g + eq^3 + \ell q^5 \right. \\ &\quad \left. + (2(e - z\lambda)(1 + xb^5 + g)^4 + \lambda b^5 + \lambda b^5 q^4)q^7 \right] (n^2 - n) \\ &\quad \left. + \left[q^3 - b\ell + \ell + \theta\lambda q^3 + (b^5 - 2)q^8 \right] (n^2 - 1), \right. \\ \eta n^2 &= \binom{2r}{r}. \end{aligned}$$

Here there are fourteen unknowns, $b, e, g, \ell, m, n, q, r, t, w, \alpha, \eta, \theta, \lambda$, and the four parameters x, z, u, y . Only one binomial coefficient and one exponential function appear. Next these are eliminated so that we [861] obtain a system of purely polynomial equations.

¹Editorial clarification: this gives a suitable effective one-parameter indexing; it need not coincide numerically with a previously chosen indexing. The displayed coding is injective on positive-integer triples.

Theorem 3. *In order that $x \in W_{(z,u,y)}$, it is necessary and sufficient that the following system of equations have a solution in positive integers.²*

$$\begin{aligned}
elg^2 + \alpha &= (b - xy)q^2, & q &= b^{560}, \\
\lambda + q^4 &= 1 + \lambda b^5, & \theta + 2z &= b^5, \\
\ell &= u + t\theta, & e &= y + m\theta, & n &= q^{16}, \\
r &= \left[g + eq^3 + \ell q^5 \right. \\
&\quad \left. + (2(e - z\lambda)(1 + xb^5 + g)^4 + \lambda b^5 + \lambda b^5 q^4)q^7 \right] (n^2 - n) \\
&\quad + \left[q^3 - b\ell + \ell + \theta\lambda q^3 + (b^5 - 2)q^8 \right] (n^2 - 1), \\
p &= 2ws^2r^2n^6, & p^2k^2 - k^2 + 1 &= \tau^2, \\
4(c - ksn^2)^2 + \eta &= k^2, \\
k &= r + 1 + hp - h, & a &= (wn^2 + 1)rsn^2, \\
c &= 2r + 1 + \varphi, & d &= bw + ca - 2c + 4a\gamma - 5\gamma, \\
d^2 &= (a^2 - 1)c^2 + 1, & f^2 &= (a^2 - 1)i^2c^4 + 1, \\
(d + of)^2 &= \left((a + f^2(d^2 - a))^2 - 1 \right) (2r + 1 + jc)^2 + 1.
\end{aligned}$$

The equations of Theorem 3 have twenty-eight unknowns,

$$a, b, c, d, e, f, g, h, i, j, k, \ell, m, n, o, p, q, r, s, t, w, \alpha, \gamma, \eta, \theta, \lambda, \tau, \varphi.$$

The degree is 5^{60} ; however, the equation $q = b^{560}$ can be replaced by certain others of low degree. In fact, by introducing some 30 additional unknowns and new equations one can reduce the degree of the system to 2. Then, by transposing terms to one side and summing squares, one can construct a universal diophantine equation in 58 unknowns and of degree 4.

Alternatively, one may try instead to reduce the total number of unknowns, ν . In [6] Julia Robinson and Ju. Matijasevič showed that ν can be reduced universally to 13. More recently Matijasevič [5] has improved this to $\nu = 9$. The corresponding value of the degree, δ , is, however, very large. The following table gives various simultaneous possibilities for δ and ν , sufficient for a universal equation.

Theorem 4. *The following universal constructions are reported. The large degrees written in scientific notation are approximate sizes, not exact integer degree bounds.*

ν	δ	ν	δ
58	4	21	96
38	8	19	2668
32	12	14	$\approx 2.0 \times 10^5$
29	16	13	$\approx 6.6 \times 10^{43}$
28	20	12	$\approx 1.3 \times 10^{44}$
26	24	11	$\approx 4.6 \times 10^{44}$
25	28	10	$\approx 8.6 \times 10^{44}$
24	36	9	D_9

²Editorial emendation: the equation for d contains $4a\gamma$, with Latin a , not $4\alpha\gamma$ as printed in the 1980 article. The latter is an original typographical error, not an OCR error; see the accompanying notes for corroboration and the Pell-equation congruence that supports the correction.

Here the nine-unknown degree is the expression reported in the later full derivation (Jones, 1982, §3):

$$D_9 = 47216 \cdot 5^{58} + 9728 \approx 1.638134073 \times 10^{45} < 1.7 \times 10^{45}.$$

The printed 1.6×10^{45} rounds this expression downward. The editorial notes (entry 1980-18) distinguish evaluation of this exact expression from a fresh derivation of the complete relation-combining degree.

A different measure of size and complexity of a system of diophantine equations is the total number [862] of indicated arithmetical operations, o , i.e. the number of additions, subtractions and multiplications necessary to evaluate or determine the correctness of a proposed solution. For the equations of Theorem 3, modified to do away with $q = b^{560}$, it can be shown that $o = 100$. This number, o , can be given an interesting interpretation as a proof-theoretic complexity bound.

Via Gödel numbering, the theorems of an effectively axiomatizable theory T become, in effect, an r.e. set, and the search for proofs becomes the search for solutions of a diophantine equation. A formal deduction is effectively recoverable from such a solution, but this recoverability does not assert preservation of proof length or computational complexity. Hence we have

Theorem 5. *For any effectively axiomatizable theory T and any proposition P , if P has a proof in T , then P has another proof consisting of 100 additions and multiplications of integers.³*

For the universal equations in [3] the value of o was 243. The bound obtained here is $o = 100$. As with ν and δ , it would be interesting to know the minimum value of o .

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³Editorial clarification: “another proof” here means an arithmetic certificate under a fixed effective encoding of provability, not necessarily a 100-step derivation in the original proof calculus of T . The count is of the indicated arithmetic operations on whole integers in the modified system; it is not a bound on the lengths of the integers, the number of bit operations, or the work of finding a certificate. The preceding operation-count definition includes subtraction. The count 100 is reproduced by counting every indicated sign of Theorem 3 once (powers not counted); that reading, and certificates under a calculator model, are examined in [jones1980_theorem5_operations.pdf](#) in this directory.